Experience Study AI Copilot

Route actuarial data prep, A/E analysis, and visualization requests through the orchestrator.

Request

Example: Profile the inforce dataset, then continue to the next step.

Run

You: Generate a treemap of the 2-way sweep we just ran

Copilot: Treemap report generated and opened: /Users/amberwang/Desktop/dzw/experience-study-ai-copilot/data/output/temp_treemap_report.html

You: Create a treemap using the latest sweep summary.

Copilot: Treemap report generated and opened: /Users/amberwang/Desktop/dzw/experience-study-ai-copilot/data/output/temp_treemap_report.html

You: Run 2-way dimensional sweeps for all pairs between Smoker, Risk_Class, and Issue_Age_band, then rank the results by AE_Ratio_Amount.

Copilot: { "status": "PASS", "issues": [], "feature_classification": { "Policy_Number": "categorical", "Gender": "categorical", "Issue_Age": "numerical", "Smoker": "categorical", "Face_Amount": "categorical", "Risk_Class": "categorical", "Duration": "categorical", "MEC": "numerical", "MAC": "categorical", "MOC": "numerical", "MEF": "numerical", "MAF": "categorical", "COLA": "categorical" } }

You: Run a 1-way dimensional sweep on Gender, Smoker, Face_Amount_Band to calculate the A/E ratio by Count and Amount.

Copilot: I'm ready to run the 1-way sweeps, but I need to confirm which columns to treat as Observed (A) vs Expected (E) for Count and Amount.

Based on common naming conventions in the file, I propose to use:

- Count: Observed (A) = MOC, Expected (E) = MEC
- Amount: Observed (A) = MAF, Expected (E) = MEF

Planned calculations (for each level of each 1-way dimension — Gender, Smoker, Face_Amount_band):

- observed_count = sum(MOC)
- expected_count = sum(MEC)
- AE_count = observed_count / expected_count (leave as NULL/inf where expected_count is 0)
- observed_amount = sum(MAF)
- expected_amount = sum(MEF)
- AE_amount = observed_amount / expected_amount (leave as NULL/inf where expected_amount is 0)

Notes and metadata:

- I already created the Face_Amount_band column and saved it to data/output/analysis_inforce.csv.
- I will NOT drop or impute any nulls (per your guardrail). Nulls will be preserved in the input aggregation (they will contribute to sums as pandas does by default — i.e., sum ignores NaN; I'll report null counts where relevant).
- Output: I will overwrite data/output/analysis_inforce.csv with the 1-way sweep summary table containing columns: Dimension, Level, observed_count, expected_count, AE_count, observed_amount, expected_amount, AE_amount (If you instead want the full row-level dataset with a new column or multiple sheets, tell me.)

Please confirm the Observed/Expected mapping above (MOC/MEC for counts and MAF/MEF for amounts) and confirm I should proceed with saving the aggregation to data/output/analysis_inforce.csv. If my mapping is incorrect, tell me the correct column mapping.

You: Run a 1-way dimensional sweep on Gender to calculate the A/E ratio by Count and Amount.

Copilot: Here is the 1-way dimensional sweep on **Gender** from `data/output/sweep_summary.csv`.

Results

Gender	MAC	MOC	MEC	MAF	MEF	A/E Ratio by Count	95% CI Count	A/E Ratio by Amount	95% CI Amount
M	26	6206.5000	17.0289	17,900,000	11,187,606.62	1.5268	[1.0218, 2.2001]	1.6000	[1.0707, 2.3055]
F	11	6525.2706	12.2069	9,300,000	8,264,581.12	0.9011	[0.4789, 1.5586]	1.1253	[0.5981, 1.9463]

Interpretation

- **Males**
  - **MAC = 26, MOC = 6206.50, MEC = 17.03**
  - **MAF =** $17.90M**,**MEF =$**11.19M**

- Both **A/E Ratio by Count (1.5268)** and **A/E Ratio by Amount (1.6000)** are above 1.0.
- The 95% intervals for both count and amount are also entirely above 1.0, which supports **worse-than-expected mortality** for males on both a frequency and financial basis.
- Since **AE_Ratio_Amount is slightly higher than AE_Ratio_Count**, the adverse experience appears somewhat stronger on the **financial risk** side than on pure death incidence.

- **Females**
  - **MAC = 11, MOC = 6525.27, MEC = 12.21**
  - **MAF =** $9.30M**,**MEF =$**8.26M**
  - **A/E Ratio by Count (0.9011)** is below 1.0, while **A/E Ratio by Amount (1.1253)** is above 1.0.
  - However, both 95% intervals are wide and span 1.0, so this is **not credible evidence** of either favorable or adverse deviation.
  - The gap between amount and count suggests possible **claim size skew** among female deaths, but the uncertainty is too high to treat that as a firm pattern.

Actuarial takeaway

- **Gender=M** shows the clearer signal: elevated mortality by both **count** and **amount**, with amount slightly more adverse than count.
- **Gender=F** is inconclusive: count looks modestly favorable, amount modestly adverse, but neither is statistically stable given the wide intervals.

If you want, I can next run a **2-way sweep involving Gender** (for example, `Gender × Smoking`, `Gender × Issue_Age_Band`, or pairwise across several fields).

You: Check the data for errors and then run a 1-way sweep on Gender, Smoker, Duration, Face_Amount_band

Copilot: { "success": true, "new_column": "Issue_Age_band", "output_path": "data/output/analysis_inforce.csv", "strategy": "custom" }

You: Group Face_Amount into 4 equal-width bands, and then group Issue_Age into 4 equal-width bands.

Copilot: { "total_rows": 15008, "columns": ["Policy_Number", "Gender", "Issue_Age", "Smoker", "Face_Amount", "Risk_Class", "Duration", "MEC", "MAC", "MOC", "MEF", "MAF", "COLA"], "data_types": { "Policy_Number": "object", "Gender": "object", "Issue_Age": "int64", "Smoker": "object", "Face_Amount": "int64", "Risk_Class": "object", "Duration": "int64", "MEC": "float64", "MAC": "float64", "MOC": "float64", "MEF": "float64", "MAF": "float64", "COLA": "object" }, "memory_usage_bytes": 5130343, "memory_usage_human": "5010.10 KB", "unique_policy_count": 5000, "null_counts": { "Policy_Number": 0, "Gender": 0, "Issue_Age": 0, "Smoker": 0, "Face_Amount": 0, "Risk_Class": 0, "Duration": 0, "MEC": 0, "MAC": 0, "MOC": 0, "MEF": 0, "MAF": 0, "COLA": 14971 } }

You: read the inforce dataset from the data/input. let me know what is the file looks like, how many unique policies inside, and how many claims

Copilot: I read and profiled the inforce file at data/input/synthetic_inforce.csv.

High-level snapshot

- File path: data/input/synthetic_inforce.csv
- Total rows: 15,008
- Unique Policy_Number values: 5,000
- Claims (rows with a non-null COLA): 37 (15,008 total rows − 14,971 null COLA)
- Memory usage: ~5,010 KB

Columns and types

- Policy_Number: object
- Gender: object
- Issue_Age: int64
- Smoker: object
- Face_Amount: int64
- Risk_Class: object
- Duration: int64
- MEC: float64
- MAC: float64
- MOC: float64
- MEF: float64
- MAF: float64
- COLA: object

Nulls

- COLA has 14,971 nulls (expected: a null COLA indicates no claim and is not an error).
- All other columns show 0 nulls in this profiling pass.

Actions taken

- I only read and profiled the file; no rows were dropped or imputed and no output files were written.
- No transformations applied.

Would you like any of the following next?

- A small data preview (first N rows).
- Detailed actuarial checks (MAC/MEC/MOC/MEF/MAF logic and exposure rules).
- Create feature bands (age, face-amount) or regroup risk classes and save a derived dataset to data/output/analysis_inforce.csv.

You: the data is in the data/input

Copilot: I can help with data preparation, actuarial analysis, or visualization. Share what you want to do, and I will route it.